QUIZ NO 1

231678

AROOBA

VISUAL PROGRAMMING

AROOBA

[Email address]

QUESTION:

Create a **C# OOP-based billing program** for PopSalon, allowing customers to select from **11 popcorn flavors** and **4 pack sizes** with correct pricing. Implement **optional gift wrapping (Rs. 50)** and **additional toppings** (Chocolate Sauce: Rs. 50, Caramel Sauce: Rs. 30, Melted Cheese: Rs. 60). Collect **customer details and feedback**, display a final bill, and include a **destructor for a thank-you message**.

```
using System;
using System.Collections.Generic;
using static PopSalon;

public class PopSalon
{
    private string customerName;
    private string mobileNumber;
    private string emailAddress;
    private string quality;
    private string flavor;
    private PackPrice packSize;
    private bool giftWrap;
    private List<Toppings> additionalToppings = new List<Toppings>();
    private double totalPrice;

    private readonly string[] flavors = {
        "Chocolate", "English Toffee", "Salted Caramel", "Caramel", "Jalapeño",
        "Cheese", "Spiced Cheese", "Plain Salted", "Buttered", "Salt and Pepper",
"Garlic"
    };

    private readonly string[] experiences = { "Good", "Neutral", "Bad" };

    public enum PackPrice
    {
        Small = 100,
        Medium = 250,
        Large = 500,
        Tin = 750
    }

    public enum Toppings
    {
        ChocolateSauce = 50,
        CaramelSauce = 30,
        MeltedCheese = 60
    }

    public PopSalon()
    {
        Console.WriteLine("Welcome to PopSalon!");
    }

    public void GetCustomerDetails()
    {
        Console.Write("Enter your name: ");
```

```
        customerName = Console.ReadLine();

        Console.Write("Enter your mobile number: ");
        mobileNumber = Console.ReadLine();

        Console.Write("Enter your email address: ");
        emailAddress = Console.ReadLine();

        Console.WriteLine("Rate your experience (1. Good, 2. Neutral, 3. Bad):");
        int expChoice;
        while (true)
        {
            if (int.TryParse(Console.ReadLine(), out expChoice) && expChoice >= 1
&& expChoice <= 3)
            {
                quality = experiences[expChoice - 1];
                break;
            }
            else
            {
                Console.WriteLine("Invalid input. Please enter 1, 2, or 3.");
            }
        }
    }

    public void SelectFlavor()
    {
        Console.WriteLine("Choose a flavor:");
        for (int i = 0; i < flavors.Length; i++)
        {
            Console.WriteLine($"{i + 1}. {flavors[i]}");
        }

        int flavorChoice;
        while (true)
        {
            if (int.TryParse(Console.ReadLine(), out flavorChoice) && flavorChoice
>= 1 && flavorChoice <= flavors.Length)
            {
                flavor = flavors[flavorChoice - 1];
                break;
            }
            else
            {
                Console.WriteLine("Invalid input. Please select a valid flavor.");
            }
        }
    }

    public void SelectPackSize()
    {
        Console.WriteLine("Select pack size:");
        PackPrice[] packSizes = (PackPrice[])Enum.GetValues(typeof(PackPrice));

        for (int i = 0; i < packSizes.Length; i++)
        {
            Console.WriteLine($"{i + 1}. {packSizes[i]} (Rs.
{(int)packSizes[i]})");
```

```
        }

        int packChoice;
        while (true)
        {
            if (int.TryParse(Console.ReadLine(), out packChoice) && packChoice >=
1 && packChoice <= packSizes.Length)
            {
                packSize = packSizes[packChoice - 1];
                totalPrice += (int)packSize;
                break;
            }
            else
            {
                Console.WriteLine("Invalid input. Please select a valid pack
size.");
            }
        }
    }

    public void CheckGiftWrap()
    {
        Console.Write("Would you like a gift wrap for Rs. 50? (yes/no): ");
        string choice = Console.ReadLine().ToLower();

        if (choice == "yes")
        {
            giftWrap = true;
            totalPrice += 50;
        }
        else
        {
            giftWrap = false;
        }
    }

    public void SelectToppings()
    {
        Console.WriteLine("Would you like additional toppings?");
        Toppings[] toppingOptions = (Toppings[])Enum.GetValues(typeof(Toppings));

        for (int i = 0; i < toppingOptions.Length; i++)
        {
            Console.WriteLine($"{i + 1}. {toppingOptions[i]} - Rs.
{(int)toppingOptions[i]}");
        }
        Console.WriteLine($"{toppingOptions.Length + 1}. No Toppings");

        while (true)
        {
            Console.Write("Enter your choice (or select the last option to
finish): ");
            int toppingChoice;
            if (int.TryParse(Console.ReadLine(), out toppingChoice) &&
toppingChoice >= 1 && toppingChoice <= toppingOptions.Length + 1)
            {
                if (toppingChoice == toppingOptions.Length + 1)
                {
```

```
                    break;
                }

                Toppings selectedTopping = toppingOptions[toppingChoice - 1];
                additionalToppings.Add(selectedTopping);
                totalPrice += (int)selectedTopping;
            }
            else
            {
                Console.WriteLine("Invalid input. Please enter a number between 1
and " + (toppingOptions.Length + 1));
            }
        }
    }

    public void DisplayBill()
    {
        Console.WriteLine("\n--- Final Bill ---");
        Console.WriteLine($"Customer Name: {customerName}");
        Console.WriteLine($"Mobile: {mobileNumber}");
        Console.WriteLine($"Email: {emailAddress}");
        Console.WriteLine($"Flavor: {flavor}");
        Console.WriteLine($"Pack Size: {packSize} (Rs. {(int)packSize})");
        Console.WriteLine($"Gift Wrap: {(giftWrap ? "Yes (Rs. 50)" : "No")}");
        Console.WriteLine($"Additional Toppings: {(additionalToppings.Count > 0 ?
string.Join(", ", additionalToppings) : "None")}");
        Console.WriteLine($"Total Price: Rs. {totalPrice}");
        Console.WriteLine($"Experience Rating: {quality}");
    }

    ~PopSalon()
    {
        Console.WriteLine("\nThank you for visiting PopSalon! We hope to see you
again.");
    }
}

class Program
{
    static void Main(string[] args)
    {
        PopSalon order = new PopSalon();
        order.SelectFlavor();
        order.SelectPackSize();
        order.CheckGiftWrap();
        order.SelectToppings();
        order.GetCustomerDetails();
        order.DisplayBill();
    }
}
```

```
Welcome to PopSalon!
Choose a flavor:
1. Chocolate
2. English Toffee
3. Salted Caramel
4. Caramel
5. Jalapeño
6. Cheese
7. Spiced Cheese
8. Plain Salted
9. Buttered
10. Salt and Pepper
11. Garlic
5
Select pack size:
1. Small (Rs. 100)
2. Medium (Rs. 250)
3. Large (Rs. 500)
4. Tin (Rs. 750)
3
Would you like a gift wrap for Rs. 50? (yes/no): yes
Would you like additional toppings?
1. CaramelSauce - Rs. 30
2. ChocolateSauce - Rs. 50
3. MeltedCheese - Rs. 60
4. No Toppings
Enter your choice (or select the last option to finish): 4
Enter your name: Arooba
Enter your mobile number: 03105539727
Enter your email address: arooba@gmail.com
Rate your experience (1. Good, 2. Neutral, 3. Bad):
1

--- Final Bill ---
Customer Name: Arooba
Mobile: 03105539727
Email: arooba@gmail.com
Flavor: Jalapeño
Pack Size: Large (Rs. 500)
Gift Wrap: Yes (Rs. 50)
Additional Toppings: None
Total Price: Rs. 550
Experience Rating: Good

Thank you for visiting PopSalon! We hope to see you again.
Press any key to continue . . .
```